Aman Kumar Bhamboo

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SUMMARY

Final year CS student with proven experience building production grade ETL pipelines, SQL cohort analytics, and multi-page Tableau dashboards on datasets exceeding 1GB. IEEE-published researcher who turns fragmented raw data into boardroom-ready insights that drive measurable business decisions.

ACHIEVEMENTS

Research Publications

"Deep Learning-Based Approaches for Efficient Brain MRI Segmentation: A Comprehensive Analysis," IEEE STCR 2025, pp. 1-9, doi: 10.1109/STCR62650.2025.11020327.

- Aggregated MRI datasets across multiple hospital sources and benchmarked U-Net, ResNet, and transformer-based architectures for brain MRI segmentation across accuracy, computational efficiency, and clinical applicability in resource-constrained environments.

PROJECTS

Corporate Revenue Leakage & Discount Audit Dashboard GitHub | Tableau Dashboard

- **Problem:** Uncontrolled sales discounting across corporate sales channels led to hidden profit erosion, making it difficult for management to identify specific leakages and track actual profitability at scale.
- Engineered an interactive Tableau financial audit dashboard to visualize corporate revenue pipelines, enabling non-technical executives to track and audit over **229 Billion in gross financial transactions**.
- Conducted a dual-axis "Financial Tug-of-War" trend analysis to mathematically isolate the impact of discount percentages (0–10%) on profit margins, proving a direct correlation with severe revenue erosion.
- Developed a granular Root-Cause Analysis matrix that mapped **11.48 Billion in discount leakage** across thousands of transactions, successfully isolating performance anomalies down to individual clerk and order levels.
- Evaluated critical corporate financial health metrics including **Gross Revenue, Discount Erosion, Net Revenue, and Clerk-wise order handling distributions** to pinpoint high-risk margin-bleeding channels.

Measuring Customer Revenue Health GitHub | Tableau Dashboard

- **Problem:** A Brazilian e-commerce marketplace had zero visibility into why customers weren't returning with $\sim$100K orders and no retention or churn metrics in place.
- Uncovered an **8% repeat purchase rate** and **77% CSAT**, with cohort analysis pinpointing $>$**70% churn within the first 30 days** recommended a 7-day post-purchase email nudge as the highest-ROI early intervention.
- Identified regional fulfillment bottlenecks by correlating seller freight duration with delivery times across all 27 states **Roraima and Amapá averaged 29+ day delays vs. São Paulo's 8 days** flagging logistics gaps as a primary churn driver.
- Cleaned and normalized **9 raw CSV files** using Python and pandas; exported to PostgreSQL and computed CRR, RPR, LTV, CSAT, and NPS using **window functions and cohort pivot queries.**
- Built a **4-page interactive Tableau dashboard** covering executive KPIs, geographic revenue breakdown, cohort retention heatmaps, and fulfillment translating raw order data into boardroom-ready insights.

EDUCATION

Vellore Institute of Technology Bhopal

Bachelor of Technology in Computer Science *(Final Year , CGPA - 7.8)* 2022–2026

TECHNICAL SKILLS

Languages & Databases: Python (pandas, NumPy, matplotlib, seaborn), SQL (PostgreSQL, MySQL) - Window Functions, CTEs, Cohort Pivot Queries, Query Optimization

Excel & Reporting: Advanced Excel (VLOOKUPs, INDEX/MATCH, Pivot Tables & Charts, Power Query, Conditional Formatting), Executive Report & Slide Preparation

BI & Visualization: Tableau (LOD Expressions, Parameter Actions, Multi-page Dashboard Design, Performance Optimization), Data-Driven Storytelling

Data Processing & Analytics: Exploratory Data Analysis (EDA), ETL Pipeline Development (YAML-driven, config-based), Data Cleaning & Validation, Spatial Clustering, Cohort & Retention Analysis, Trend & Pattern Analysis

Core Competencies: Business Metric Design (CRR, LTV, NPS, CSAT), Large-scale Data Processing (3.1 M+), Hypothesis-Driven Analysis